

A Limit-Ordinal Partial Answer for Subspaces Whose Weak-Star Derived Sets Are Proper and Norm Dense

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Status

This packet records a likely valid partial result for the limit-ordinal question at the end of Silber's paper [1]. It proves the desired conclusion whenever X contains a non-quasi-reflexive subspace with separable dual. The general case from Silber's question remains open.

Source Question

The source paper is Zdenek Silber, *On subspaces whose weak* derived sets are proper and norm dense*, arXiv:2203.00288 [1]. After proving the successor-ordinal theorem, Silber explains that the proof does not cover limit ordinals and asks the following question on page 11.

$E : \dot{W} \rightarrow X$ be the identity embedding. Then $E^* : X^* \rightarrow \dot{W}^*$ is the restriction map. Set $A = (E^*)^{-1}(K)$. It follows from [11, Lemma 1] and the fact that E^* is onto that $A^{(\alpha)} = (E^*)^{-1}(K^{(\alpha)}) \subsetneq X^*$. As E^* is an open mapping, the preimage of a dense set is dense. Thus $\overline{A^{(\alpha)}} = X^*$. $\square$

Note that in Theorem 2.1 (3) we restrict ourselves only to successor ordinals. The reason lies in the proof of Claim 3 of Theorem 2.10. More specifically, we needed to pass from a general $y^* \in U^\perp$ to elements $y_n^* = P_n^*(y^*)$ with finite support. Following the proof of Claim 3 for limit ordinal α , with considering $(\alpha_n)_{n=1}^\infty$ instead of $\alpha - 1$ (see Construction 2.4), we would end with $z_{n_k}^* \in K^{(\alpha_{n_k})}$, and thus $z^* \in (\bigcup_{k=1}^\infty K^{(\alpha_{n_k})})^{(1)} \subseteq K^{(\alpha+1)}$. We would, however, need z^* to be in $K^{(\alpha)}$. The problem for limit ordinals thus remains open:

Question 1. *Let α be a limit ordinal and X be a non-quasi-reflexive Banach space containing an infinite-dimensional subspace with separable dual. Is there a subspace A of X^* such that $A^{(\alpha)} \subsetneq \overline{A^{(\alpha)}} = X^*$?*

REFERENCES

- [1] ALBIAC, F., AND KALTON, N. *Topics in Banach Space Theory*. No. sv. 10 in Graduate Texts in Mathematics. Springer, 2006.
- [2] BANACH, S. *Théorie des opérations linéaires*. Éditions Jacques Gabay, Sceaux, 1993. Reprint of the 1932 original.

In the notation of [1], if $A \subset X^*$, then

$$A^{(1)} = \bigcup_{n=1}^{\infty} \overline{A \cap nB_{X^*}}^{w^*},$$

successor derivatives are defined by $A^{(\beta+1)} = (A^{(\beta)})^{(1)}$, and limit derivatives by $A^{(\lambda)} = \bigcup_{\beta < \lambda} A^{(\beta)}$.

Proof Intuition

Silber's recursive pieces $K(\alpha, A, z + af_k)$ are already defined for limit ordinals. If one repeats the successor proof at a limit ordinal λ , the finite-support approximants land in cofinally increasing stages $K^{(\lambda_n)}$, so their final weak-star limit is only known to belong to $K^{(\lambda+1)}$. This is the obstruction explicitly identified in the source paper.

The obstruction disappears when the relevant finite-dimensional decomposition is shrinking: then $P_n^* y^* \rightarrow y^*$ in norm, not merely weak-star. In a non-quasi-reflexive space with separable dual, the Davis-Johnson/Ostrovskii minimal-system construction can be run with a dense sequence in the whole dual. The same annihilation bookkeeping used to prove the u_n -basis is shrinking then proves the whole FDD is shrinking. This norm convergence makes the limit-stage approximants norm dense. Properness is kept by the cofinal sequence $\lambda_n \nearrow \lambda$, which forces every element of $K^{(\lambda)}$ to satisfy all but finitely many tail equations.

A Shrinking-FDD Variant of Silber's Setup

Lemma 1. *Let Y be a non-quasi-reflexive Banach space with separable dual. Then Silber's Lemma 2.2 may be realized inside Y so that the finite-dimensional decomposition*

$$[x_1, x_2, u_1], [x_3, x_4, x_5, u_2], \dots, [x_{n_p}, x_{n_p+1}, \dots, x_{n_p+p}, u_p], \dots$$

of $W = \overline{\text{span}}\{x_i, u_i : i \in \mathbb{N}\}$ is shrinking.

Proof. The proof is the construction behind [1, Lemma 2.2], which is taken from [3, Lemma 2]. In Ostrovskii's proof, one first chooses a sequence $(s_i) \subset X^*$ whose restrictions are dense in the dual of the auxiliary separable-dual subspace, and one constructs increasing finite-dimensional spaces $G_i \subset X^*$ with $s_i \in G_i$. Future vectors u_j and x_{n_j+k} are chosen in $G_{j-1}^\perp$.

When the ambient space is Y and Y^* is separable, choose (s_i) norm dense in the whole Y^* . The Davis-Johnson local-reflexivity construction is otherwise unchanged and still gives Silber's uniformly bounded system, the bounded row partial sums, and the displayed FDD.

Let $E_p = [x_{n_p}, x_{n_p+1}, \dots, x_{n_p+p}, u_p]$. We verify that (E_p) is shrinking. Take $w^* \in W^*$, extend it by Hahn-Banach to $\tilde{w}^* \in Y^*$, and fix $\varepsilon > 0$. Choose m with $\|\tilde{w}^* - s_m\| < \varepsilon$. If $p > m$, then

$$E_p \subset G_{p-1}^\perp \subset G_m^\perp \subset \ker s_m.$$

Hence for $z \in \overline{\text{span}}\{E_p : p > m\}$,

$$|w^*(z)| = |(\tilde{w}^* - s_m)(z)| \leq \varepsilon \|z\|.$$

Thus the norm of w^* on the m -tail is at most ε . This is the standard tail criterion for a shrinking FDD. Therefore $P_m^* w^* \rightarrow w^*$ in norm for the associated partial-sum projections. $\square$

Theorem

Theorem 1. *Let $\lambda < \omega_1$ be a countable limit ordinal. Suppose a Banach space X contains a subspace Y such that Y is non-quasi-reflexive and Y^* is separable. Then there is a linear subspace $A \subset X^*$ such that*

$$A^{(\lambda)} \subsetneq \overline{A^{(\lambda)}}^{\|\cdot\|} = X^*.$$

Proof. It is enough to prove the assertion for the subspace W produced by Lemma 1. Indeed, if $E : W \rightarrow X$ is the inclusion and $K \subset W^*$ has $K^{(\lambda)}$ proper and norm dense in W^* , set

$$A = (E^*)^{-1}(K) \subset X^*.$$

The restriction map $E^* : X^* \rightarrow W^*$ is onto. By the standard lifting lemma used in [2, Lemma 1] and in the proof of [1, Theorem 2.1],

$$A^{(\lambda)} = (E^*)^{-1}(K^{(\lambda)}).$$

The preimage of a proper subset under an onto map is proper, and the preimage of a norm dense set under an open onto operator is norm dense.

We now work in W^* . Use the notation from [1]: for each row k , let $f_k \in W^{**}$ be a weak-star cluster point of the bounded partial sums $\sum_{j=1}^m x_j^k$. The identities

$$f_k(\tilde{x}_s^t) = \delta_{k,t}, \quad f_k(\tilde{u}_n) = 0$$

hold by [1, Lemma 2.7]. Let P_m be the FDD partial-sum projections. By Lemma 1,

$$P_m^* y^* \rightarrow y^* \quad \text{in norm for every } y^* \in W^*.$$

Choose an increasing sequence (λ_n) of ordinals cofinal in λ , a partition $(A_n)_{n=0}^\infty$ of $\mathbb{N}$ into infinite sets, and a summable sequence $a_n > 0$. Let $q(n)$ be the increasing enumeration of A_0 . Using Silber's Construction 2.4, set

$$K_n = K(\lambda_n, A_n, u_n + a_n f_{q(n)}) \subset W^*, \quad K = \bigcap_{n=1}^\infty K_n.$$

First, $K^{(\lambda)}$ is proper. If $\beta < \lambda$, then there is N such that $\beta \leq \lambda_n$ for all $n \geq N$. Since $K \subset K_n$, monotonicity of derived sets and [1, Proposition 2.5] give

$$K^{(\beta)} \subseteq \bigcap_{n \geq N} \ker(u_n + a_n f_{q(n)}).$$

Hence

$$K^{(\lambda)} \subseteq \bigcup_{N=1}^\infty \bigcap_{n \geq N} \ker(u_n + a_n f_{q(n)}).$$

The biorthogonal functionals $\tilde{u}_n$ are uniformly bounded, so for any scalar sequence $b = (b_n) \in \ell_1$ with all $b_n \neq 0$, the series $y^* = \sum_n b_n \tilde{u}_n$ converges in W^* . Since $f_{q(n)}(\tilde{u}_m) = 0$,

$$(u_n + a_n f_{q(n)})(y^*) = b_n \neq 0$$

for every n . Thus $y^* \notin K^{(\lambda)}$.

It remains to prove norm density. First, for each n , [1, Lemma 2.8(a)] applied to $(\lambda_n, A_n, u_n + a_n f_{q(n)})$ gives

$$\tilde{u}_n \in K^{(\lambda_n+1)} \subset K^{(\lambda)},$$

because $\lambda_n + 1 < \lambda$. The inclusion into the derived set of the intersection uses the same disjoint-support verification as Claim 2 in [1, Theorem 2.10]: the relevant Q_1 -piece for the n -th coordinate is contained in every K_j , $j \neq n$.

Let $U = [u_n] \subset W$, and take $y^* \in U^\perp$. Define $y_m^* = P_m^* y^*$. Then $y_m^* \rightarrow y^*$ in norm, and y_m^* is finitely supported. As in Claim 3 of [1, Theorem 2.10], put

$$z_m^* = y_m^* - \sum_{i=1}^{\infty} a_i f_{q(i)}(y_m^*) \tilde{u}_i.$$

For each fixed m , this sum is finite. The finite-support decomposition from Silber's Claim 3 and [1, Lemma 2.8(b)], with λ_i in place of the common successor predecessor, gives

$$z_m^* \in K^{(\lambda)}.$$

Indeed each local component lies in $(K_i)^{(\lambda_i)} \subset (K_i)^{(\lambda)}$, and the Q -piece disjointness puts it in the corresponding derivative of the full intersection.

Since $P_m^* y^* \rightarrow y^*$ in norm and the functionals $f_{q(i)}$ and $\tilde{u}_i$ are uniformly bounded while $\sum_i a_i < \infty$, the correction terms converge in norm:

$$\sum_i a_i f_{q(i)}(y_m^*) \tilde{u}_i \longrightarrow \sum_i a_i f_{q(i)}(y^*) \tilde{u}_i.$$

Thus

$$z_m^* \longrightarrow z^* := y^* - \sum_i a_i f_{q(i)}(y^*) \tilde{u}_i \quad \text{in norm.}$$

So $z^* \in \overline{K^{(\lambda)}}^{\|\cdot\|}$. Since each $\tilde{u}_i$ lies in $K^{(\lambda)}$, the norm-convergent series $\sum_i a_i f_{q(i)}(y^*) \tilde{u}_i$ also belongs to $\overline{K^{(\lambda)}}^{\|\cdot\|}$. Therefore $y^* \in \overline{K^{(\lambda)}}^{\|\cdot\|}$.

Finally, let $w^* \in W^*$. Because (u_n) is shrinking, the restriction $w^*|_U \in U^*$ is norm approximable by finite linear combinations of $\tilde{u}_n|_U$. Given $\varepsilon > 0$, choose such a finite combination $u^* = \sum_{j=1}^m \lambda_j \tilde{u}_j|_U$ with $\|w^*|_U - u^*\| \leq \varepsilon$, and extend $w^*|_U - u^*$ to $r^* \in W^*$ with $\|r^*\| \leq \varepsilon$. Then

$$w^* - r^* - \sum_{j=1}^m \lambda_j \tilde{u}_j \in U^\perp \subseteq \overline{K^{(\lambda)}}^{\|\cdot\|}.$$

The finite sum belongs to $K^{(\lambda)}$, so $\text{dist}(w^*, \overline{K^{(\lambda)}}^{\|\cdot\|}) \leq \varepsilon$. Since ε is arbitrary, $\overline{K^{(\lambda)}}^{\|\cdot\|} = W^*$. $\square$

Verification Notes

The proof is a direct modification of Silber's proof of Theorem 2.10. The new ingredient is Lemma 1: when the ambient non-quasi-reflexive space has separable dual, the dense sequence used in the Davis-Johnson/Ostrovskii construction can be made dense in the whole dual, which makes the constructed FDD shrinking. This gives the norm convergence of P_m^* needed at the limit stage.

The proof deliberately does not claim this shrinking-FDD property for Ostrovskii's full general hypothesis, where the non-quasi-reflexive part and the separable-dual subspace may be different. In that setting Silber only asserts weak-star convergence of P_m^* , and the present argument does not bridge that gap.

Novelty And Literature Check

Local run indexes were searched for 2203.00288, `weak* derived`, `proper norm dense`, and `limit ordinal`; no prior packet for this question was found before the previous c_0 subcase packet from this lane. Web searches for exact and close variants of Silber’s Question 1 found Silber’s paper [1], Ostrovskii’s transfinite subspace-order paper [2], Ostrovskii’s first-derivative norm-dense paper [3], and Ostrovskii’s convex successor-order paper [4], but no later explicit solution of the limit-ordinal subspace question. This packet should therefore be reviewed as a new partial result, not as a literature-status identification.

Limitations And Human Review Recommendation

The result covers spaces containing a non-quasi-reflexive subspace with separable dual. It does not settle the remaining case of Silber’s question where X is non-quasi-reflexive and contains some infinite-dimensional subspace with separable dual, but the available separable-dual subspaces are not known to be non-quasi-reflexive. A reviewer should focus on Lemma 1 and on the imported finite-support decomposition from Claim 3 of [1, Theorem 2.10].

References

- [1] Z. Silber, *On subspaces whose weak* derived sets are proper and norm dense*, arXiv:2203.00288, 2022.
- [2] M. I. Ostrovskii, *W^* -derived sets of transfinite order of subspaces of dual Banach spaces*, arXiv:math/9303203; originally Dokl. Akad. Nauk Ukrain. SSR, 1987, no. 10, 9–12.
- [3] M. I. Ostrovskii, *Weak* closures and derived sets in dual Banach spaces*, Note Mat. 31 (2011), 129–138; arXiv:1003.5176.
- [4] M. I. Ostrovskii, *Weak* closures and derived sets for convex sets in dual Banach spaces*, arXiv:2112.04670, 2021.